

supervisor	
signature	

* Your answer should be written in English.

(f) Explain what the Amdahl's law indicates (The main conclusion of Amdahl's law)? Your answer should be short and to the point. ([use less than 20 English words.])

3. (15 points) Answer to the following questions that are related to 'lock granularity'.

(a) What is 'lock granularity'? Explain. Your answer should be short and to the point.

([use less than 15 English words.])

(b) What is the meaning of 'coarse lock granularity' ? Explain. Your answer should be short and to the point.

([use less than 10 English words.])

(c) Too-coarse lock granularity may result in poor performance. Why? Explain. Your answer should be short and to the point.

([use less than 10 English words.])

(d) What is the meaning of 'fine lock granularity' ? Explain. Your answer should be short and to the point.

([use less than 10 English words.])

(e) Too-fine lock granularity may also result in poor performance. Why? Explain. Your answer should be short and to the point.

([use less than 10 English words.])

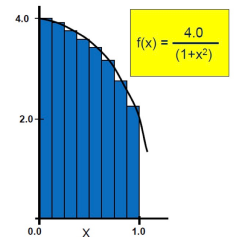
4. (33 points) Following multi-threaded JAVA code efficiently computes and displays the result of numerical

integration $\int_0^1 \frac{4.0}{(1+x^2)} dx$. Fill in empty boxes below with appropriate JAVA codes.

-----< source code : ex3.java >-----

```
class IntThread  {
    int my_id;
    int num_steps, int num_threads;
    double sum;

    IntThread(int id, int numSteps, int numThreads) {
        my_id=id; num_steps=numSteps; num_threads=numThreads;
        sum=0.0;
    }
}
```



```
class ex3 {
    private static int NUM_THREAD = 4;           // number of threads to use
    private static int NUM_STEP = 10000000;      // number of steps (rectangles to create)

    public static void main(String[] args) {
        int i; double sum=0.0;
        

        System.out.println("integration result=" + sum + "\n");
    }
}
```

-----< output result of this program >-----

integration result=3.1415926535896697